

$$r(u, v) = (2 \cosh u \cos v, 2 \cosh u \sin v, 4 \sinh u) \quad u \in \mathbb{R} \quad v \in [0, 2\pi)$$

(a) $(0, \frac{\pi}{2})$ 处的切空间

$$r_u(u, v) = (2 \sinh u \cos v, 2 \sinh u \sin v, 4 \cosh u)$$

$$r_v(u, v) = (-2 \cosh u \sin v, 2 \cosh u \cos v, 0)$$

$$r_u(0, \frac{\pi}{2}) = (0, 0, 4) \quad r_v(0, \frac{\pi}{2}) = (-2, 0, 0)$$

$$T_{(0, \frac{\pi}{2})}(r(u, v)) = \{ k_1 r_u + k_2 r_v = (-2k_2, 0, 4k_1) \mid k_1, k_2 \in \mathbb{R} \}$$

(b). $\vec{v} = a r_u + b r_v$

$$\vec{v}(0) = \alpha r_u + \beta r_v$$

由 $\frac{D\vec{v}}{dt} \equiv 0$ 得

$$(\ddot{a} + \Gamma_{11}^1 \dot{a} \dot{u} + \Gamma_{12}^1 (a \dot{v} + b \dot{u}) + \Gamma_{21}^1 b \dot{v}) r_u$$

$$+ (\ddot{b} + \Gamma_{11}^2 \dot{a} \dot{u} + \Gamma_{12}^2 (a \dot{v} + b \dot{u}) + \Gamma_{22}^2 b \dot{v}) r_v = 0$$

$$\dot{u} = 1 \quad \dot{v} = 0$$

$$(\ddot{a} + \Gamma_{11}^1 \dot{a} + \Gamma_{12}^1 b) r_u + (\ddot{b} + \Gamma_{11}^2 \dot{a} + \Gamma_{12}^2 b) r_v = 0$$

$$r_u \cdot r_v = 0$$

$$\begin{cases} \ddot{a} + \Gamma_{11}^1 \dot{a} + \Gamma_{12}^1 b = 0 \\ \ddot{b} + \Gamma_{11}^2 \dot{a} + \Gamma_{12}^2 b = 0 \end{cases} \quad (*)$$

$$\text{在 } (u, \frac{\pi}{2}) \text{ 处 } \widetilde{r}_u = r_u(u, \frac{\pi}{2}) = (0, 2 \sinh u, 4 \cosh u)$$

$$\widetilde{r}_v = r_v(u, \frac{\pi}{2}) = (-2 \cosh u, 0, 0)$$

$$\widetilde{E} = E(u, \frac{\pi}{2}) = \widetilde{r}_u \cdot \widetilde{r}_u = 4 \sinh^2 u + 16 \cosh^2 u$$

$$\widetilde{F} = F(u, \frac{\pi}{2}) = \widetilde{r}_u \cdot \widetilde{r}_v = 0$$

$$\widetilde{G} = G(u, \frac{\pi}{2}) = \widetilde{r}_v \cdot \widetilde{r}_v = 4 \cosh^2 u$$

$$r_{uu} = (2 \cosh u \cos v, 2 \cosh u \sin v, 4 \sinh u)$$

$$r_{uv} = (-2 \sinh u \sin v, 2 \sinh u \cos v, 0)$$

$$r_{vv} = (-2 \cosh u \cos v, -2 \cosh u \sin v, 0)$$

$$\widetilde{r}_{uu} = r_{uu}(u, \frac{\pi}{2}) = (0, 2 \cosh u, 4 \sinh u)$$

$$\widetilde{r}_{uv} = r_{uv}(u, \frac{\pi}{2}) = (-2 \sinh u, 0, 0)$$

$$\widetilde{r}_{vv} = r_{vv}(u, \frac{\pi}{2}) = (0, -2 \cosh u, 0)$$

$$P'_{11} = \frac{\widetilde{G} \widetilde{r}_{uu} \widetilde{r}_u - \widetilde{F} \widetilde{r}_{uu} \widetilde{r}_v}{\widetilde{E} \widetilde{G}} = \frac{5 \sinh u \cosh u}{5 \sinh^2 u + 4}$$

$$P'_{12} = \frac{\widetilde{G} \widetilde{r}_{uv} \widetilde{r}_u - \widetilde{F} \widetilde{r}_{uu} \widetilde{r}_v}{\widetilde{E} \widetilde{G}} = 0$$

$$P''_{11} = \frac{\widetilde{E} \widetilde{r}_{uu} \widetilde{r}_v - \widetilde{F} \widetilde{r}_{uu} \widetilde{r}_u}{\widetilde{E} \widetilde{G}} = 0$$

$$P''_{12} = \frac{\widetilde{E} \widetilde{r}_{uv} \widetilde{r}_v - \widetilde{F} \widetilde{r}_{uv} \widetilde{r}_u}{\widetilde{E} \widetilde{G}} = \frac{\sinh u}{\cosh u}$$

$$(*) \Rightarrow \begin{cases} \ddot{a} + \frac{5 \sinh u \cosh u}{5 \sinh^2 u + 4} a = 0 \\ \ddot{b} + \frac{\sinh u}{\cosh u} b = 0 \end{cases}$$

$$a = \frac{c}{\sqrt{5 \sinh^2 u + 4}}, \quad b = \frac{1}{c \cosh u}$$

$$a(0) = \alpha \quad c = 2\alpha$$

$$b(0) = \beta \quad c = \frac{1}{\beta}$$

$$a = \frac{2\alpha}{\sqrt{5 \sinh^2 u + 4}}$$

$$b = \frac{\beta}{\cosh u}$$

$$\therefore \vec{v} = \frac{2\alpha}{\sqrt{5 \sinh^2 u + 4}} r_u + \frac{\beta}{\cosh u} r_v$$

2. 沿纬圆 $u=2$ 的平行移动有 Holonomy 现象

recall: 沿着正则曲面上分段正则闭曲线对一个切向量作平行移动不会得到原来的切向量.

goal: $v = a r_u + b r_v$ 证明 $a(0) \neq a(2\pi)$ $b(0) \neq b(2\pi)$

$$(\dot{a} + \Gamma_{11}^1 a \dot{u} + \Gamma_{12}^1 (a \dot{u} + b \dot{v}) + \Gamma_{21}^1 b \dot{u}) r_u + (\dot{b} + \Gamma_{11}^2 a \dot{u} + \Gamma_{12}^2 (a \dot{u} + b \dot{v}) + \Gamma_{22}^2 b \dot{v}) r_v = 0$$

$$\dot{u} = 0 \quad \dot{v} = 1$$

$$\begin{cases} \dot{a} + \Gamma_{12}^1 a + \Gamma_{22}^1 b = 0 \\ \dot{b} + \Gamma_{12}^2 a + \Gamma_{22}^2 b = 0 \end{cases}$$

$$\tilde{r}_u = r_u(z, v) = (z \sinh z \cos v, z \sinh z \sin v, 4 \cosh z)$$

$$\tilde{r}_v = r_v(z, v) = (-2 \cosh z \sin v, 2 \cosh z \cos v, 0)$$

$$\tilde{r}_{uu} = r_{uu}(z, v) = (2 \cosh z \cos v, 2 \cosh z \sin v, 4 \sinh z)$$

$$\tilde{r}_{uv} = r_{uv}(z, v) = (-2 \sinh z \sin v, 2 \sinh z \cos v, 0)$$

$$\tilde{r}_{vv} = r_{vv}(z, v) = (-2 \cosh z \cos v, -2 \cosh z \sin v, 0)$$

$$\widetilde{E} = \widetilde{r}_u \cdot \widetilde{r}_u = 4 \sinh^2 z + 16 \cosh^2 z$$

$$\widetilde{F} = 0$$

$$\widetilde{G} = \widetilde{r}_v \cdot \widetilde{r}_v = 4 \cosh^2 z$$

$$\vec{v} = a \widetilde{r}_u + b \widetilde{r}_v$$

$$P'_{12} \quad P'_{22} \quad P''_{12} \quad P''_{22}$$

$$P'_{12} = \frac{\widetilde{G} \widetilde{r}_{uv} \widetilde{r}_u - \widetilde{F} \widetilde{r}_{uv} \widetilde{r}_v}{\widetilde{E} \widetilde{G}} = 0$$

$$P'_{22} = \frac{\widetilde{G} \widetilde{r}_{vv} \widetilde{r}_u - \widetilde{F} \widetilde{r}_{vv} \widetilde{r}_v}{\widetilde{E} \widetilde{G}} = - \frac{\sinh z \cosh z}{5 \sinh^2 z + 4}$$

$$P''_{12} = \frac{\widetilde{E} \widetilde{r}_{uv} \widetilde{r}_v - \widetilde{F} \widetilde{r}_{uv} \widetilde{r}_u}{\widetilde{E} \widetilde{G}} = 0$$

$$P''_{22} = \frac{\widetilde{E} \widetilde{r}_{vv} \widetilde{r}_v - \widetilde{F} \widetilde{r}_{vv} \widetilde{r}_u}{\widetilde{E} \widetilde{G}} = 0$$

$$\begin{cases} a + \frac{-\sinh z \cosh z}{5 \sinh^2 z + 4} b = 0 \\ b + \frac{\sinh z}{\cosh z} a = 0 \end{cases}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 & \frac{\sinh z \cosh z}{5 \sinh^2 z + 4} \\ \frac{\sinh z}{\cosh z} & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$$

$$a = \widetilde{C}_1 \cos \frac{\sinh z}{\sqrt{5 \sinh^2 z + 4}} v + \widetilde{C}_2 \sin \frac{\sinh z}{\sqrt{5 \sinh^2 z + 4}} v$$

$$\frac{\pi}{2} a(0) = a(2\pi)$$

$$\widetilde{C}_1 = \widetilde{C}_1 \cos \frac{\sinh z}{\sqrt{5 \sinh^2 z + 4}} 2\pi + \widetilde{C}_2 \sin \frac{\sinh z}{\sqrt{5 \sinh^2 z + 4}} 2\pi \quad (**)$$

$\therefore \widetilde{C}_1, \widetilde{C}_2$ 任意的

$$0 \leq \frac{\sinh z}{\sqrt{5 \sinh^2 z + 4}} \leq 1.$$

与(**)矛盾

$$\therefore a(0) \neq a(2\pi).$$

$b(0) \neq b(2\pi)$ 同理可证

$\therefore$ 有 Holonomy.